Figure 8-60 shows the first four structural modes for a modern sporting motorcycle. The first mode shows lateral bending of both front fork and swingarm in the same direction. The second mode shows a bending deformation of the rear part of the chassis and of the fork in the same direction; there is also a remarkable in plane and lateral deformation of the swingarm. The third and the fourth modes are more complicated and involve bending and torsion of the swingarm, of the fork and of the rear part of the chassis while the main part of the chassis has smaller deformations.

8.12 Rigid body properties and Mozzi axis

If we consider the motorcycle as a single rigid body moving in space, subject to translation in the ground plane and rotations about its roll and yaw axes, a number of interesting observations can be made.

Motorcycle motion depends on the kind of maneuver and on the riding style. Every maneuver starts with a variation of the front lateral force generated by the steering motion. If a rightward impulsive force is applied to the front tire contact point, the motorcycle moves with the following leaning and yaw velocity:

$$\dot{\varphi} = -\frac{I_{z_G} h - I_{xz_G}(p-b)}{I_{x_G} I_{z_G} - I_{xz_G}^2} \qquad \dot{\psi} = \frac{I_{x_G}(p-b) - I_{xz_G} h}{I_{x_G} I_{z_G} - I_{xz_G}^2}$$

The front wheel contact point lateral velocity is:

$$\dot{y} = -\frac{(I_{x_G} I_{z_G} - I_{xz_G}^2) - I_{x_G} mb(p-b) + mh(I_{z_G} h + I_{xz_G} b) - I_{xz_G} mh(p-b)}{(I_{x_G} I_{z_G} - I_{xz_G}^2)m}$$

The roll velocity is negative (leftward motion) while the yawing velocity is positive (rightward motion). The roll velocity increases if the roll moment of inertia I_{x_G} is decreased, while the yawing velocity increases, if the yaw moment of inertia I_{z_G} is decreased. Both velocities increase if the corresponding products of inertia (in the inertia tensor) become more negative.

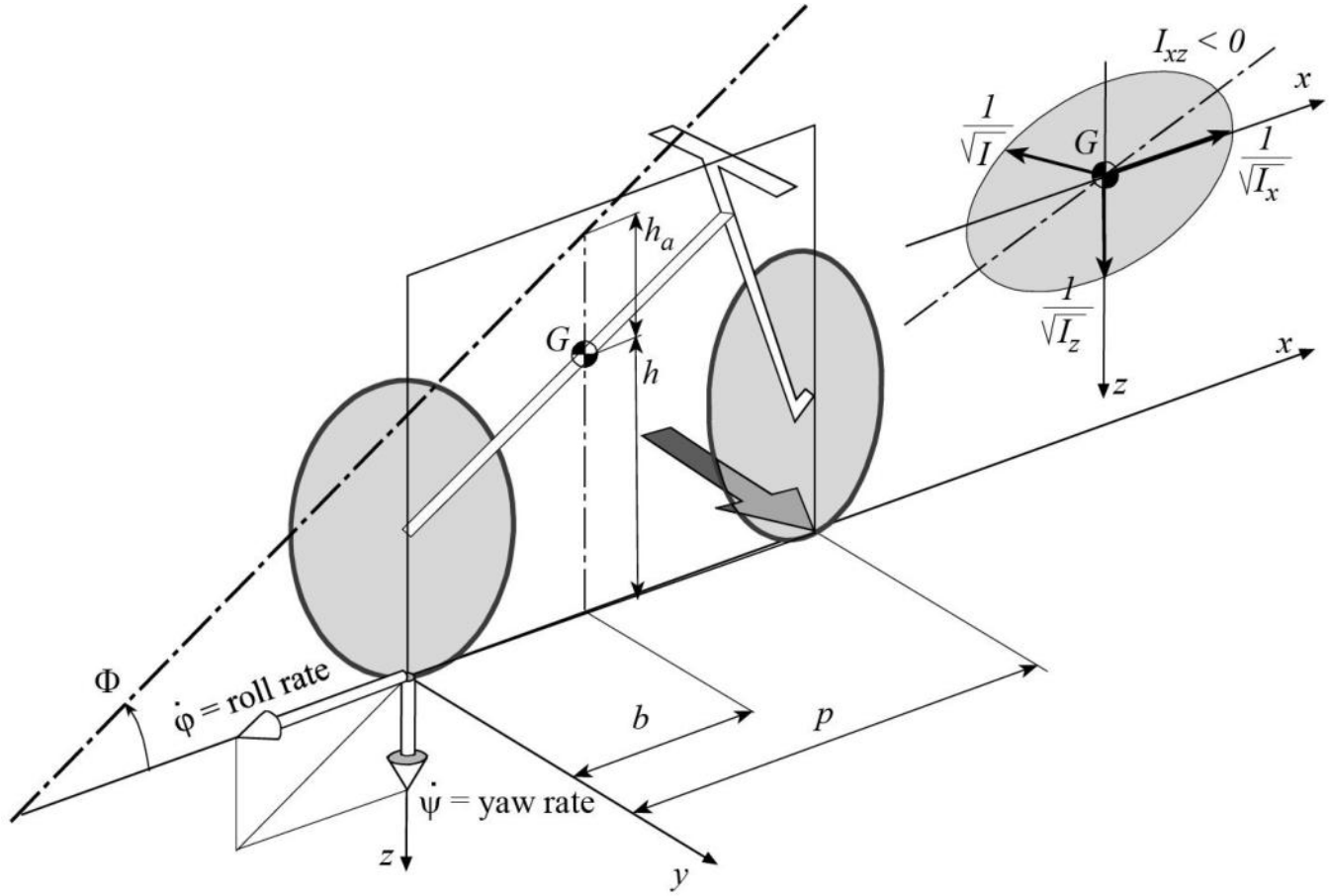

***Fig. 8-61* Instantaneous rotation axis.**

Knowing the forward velocity, the yaw velocity, and the roll velocity of motorcycle one can define a vector quantity know as the Mozzi axis, or the instantaneous axis of rotation for the rigid body. The Mozzi axis can be used to describe different transient maneuvers made by the motorcycle.

The Mozzi axis (Fig. 8-61) is described in the SAE coordinate system by:

$$y = \frac{\dot{\psi} V}{\dot{\psi}^2 + \dot{\varphi}^2} \qquad z = \frac{\dot{y} + \dot{\psi} x}{\dot{\varphi}}$$

and x is coincident with the body's center of mass location. The angle that the axis makes with the ground plane is:

$$\Phi = \arctan\left(\frac{\dot{\Psi}}{\dot{\varphi}}\right)$$

In terms of the rigid body properties the slope of the instantaneous axis of rotation with respect to the road plane *xy* is:

$$\Phi = \arctan \frac{I_{x_G}(p-b) - I_{xz_G} h}{I_{z_G} h - I_{xz_G}(p-b)}$$

The angle increases, increasing the roll inertia and decreasing the yaw one (i.e. less roll motion and more yaw).

The vertical distance of the axis from the motorcycle center of mass increases, as the roll inertia increases and product of inertia terms decrease.

People involved in motorcycle design and racing often ask themselves the following questions:

- Where is the instantaneous axis of the vehicle in motion?
- How do yaw and roll rate combine?

Answers to these questions may be found in a mathematical way by making use of the concept of the Mozzi axis. The plots derived from the Mozzi axis theory are useful in highlighting the effects of variations in path, vehicle properties, and riding style.

The geometric loci of the Mozzi trace (the point where the axis intersects the ground plane) and that of the turn center during a slalom maneuver are presented in Fig. 8-62.

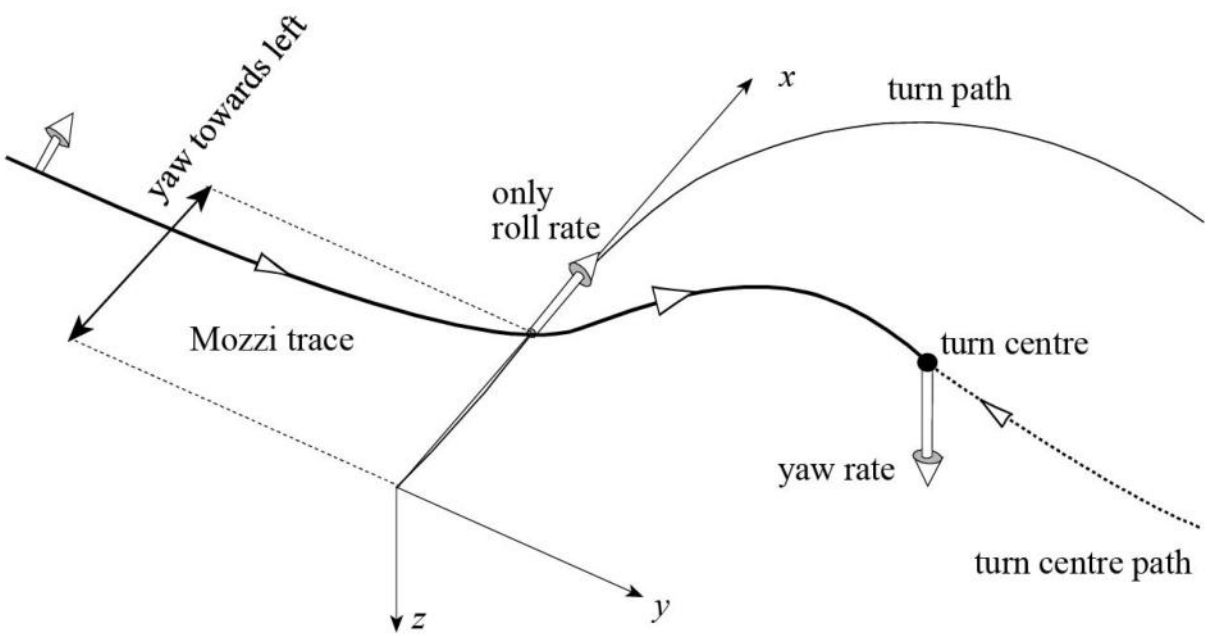

***Fig. 8-62** Mozzi trace in a slalom maneuver.*

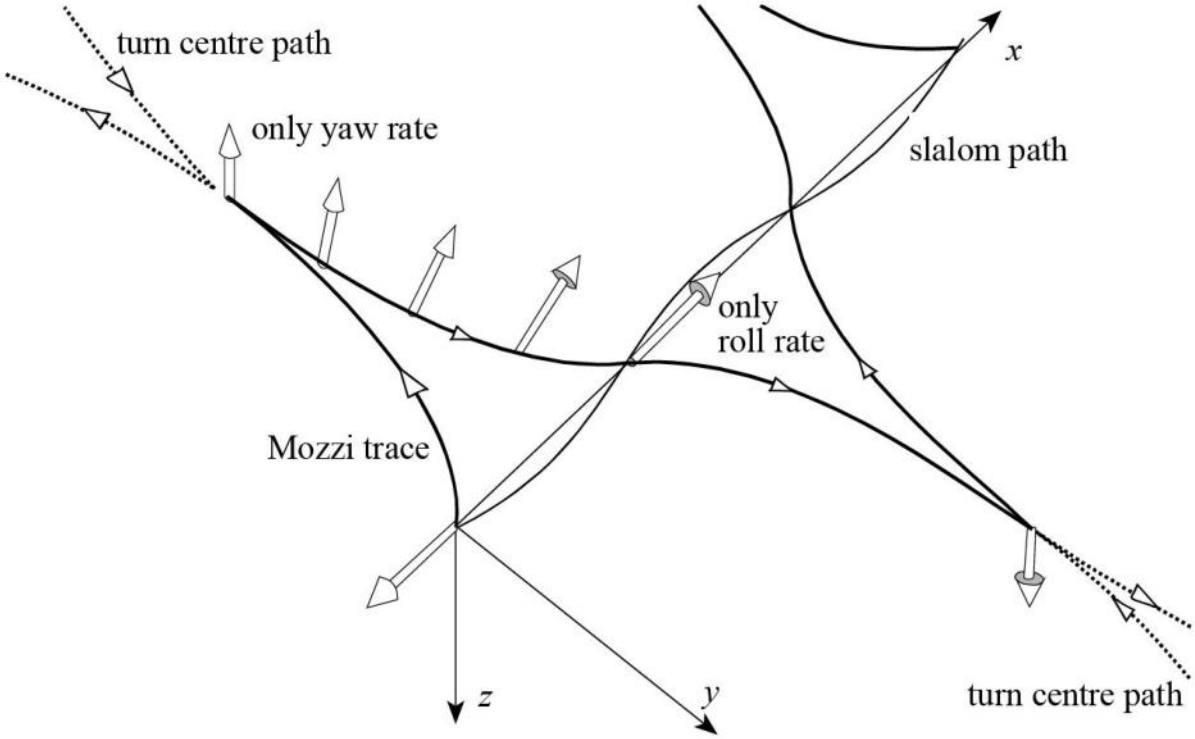

***Fig. 8-63** Mozzi trace in a entering turn maneuver.*

The locus of the turn center is close to a piecewise linear. It is worth highlighting that the turn center coordinates tend to infinity if the yaw velocity tends to zero. In the sinusoidal slalom this condition happens when the trajectory crosses the x axis.

The locus of the Mozzi trace is a curve with periodic cusps, which always lie on inner side of the path. The loci of the Mozzi trace and the turn center show periodic intersection points that take place when the radius of curvature is close to the minimum value.

Figure 8.63 deals with a motorcycle that is entering a right curve with a counter-steer technique. In this maneuver the Mozzi axis moves from the outside of the path to the center of the curve.

The yaw rate is negative (counter-yaw) at the beginning of the maneuver and positive during the rest of the maneuver. The roll rate reaches a maximum after the beginning of the rotation towards the right and then tends to zero. The Mozzi trace comes from -∞, because the path is almost straight and the yaw rate is negative, then it crosses the path when the yaw rate is zero. Finally, in the steady turning portion of the path, the Mozzi trace tends to coincide with the turn center, because the roll rate is close to zero.

8.13 Dynamic analysis with multi-body codes

Nowadays “multi-body” codes make possible the precise and complete dynamic analysis of a vehicle’s operation on the road through the use of computer simulation. MSA *(Multi-body System Analysis)* represents the computer study of movements in mechanical systems as a result of the application of external forces or stresses that act on the system. The spatial systems forming the subject of the study are simulated with rigid bodies and/or flexible elements connected to each other with various types of kinematic and dynamic connections. The external forces and resulting reactions lead to movements of the system’s components that satisfy constraint conditions. The MSA codes play a role whose importance is only destined to increase through the use of modern integrated computer-aided design. They make it possible to evaluate and optimize the characteristics and performance of a